

A sequence of seven hand-drawn arrows on a black background, illustrating the evolution of a drawing technique. The arrows are arranged horizontally from left to right. The first arrow is a simple horizontal line. The second and third arrows are identical to the first. The fourth arrow has a small vertical line (tail) at its left end. The fifth arrow has a small vertical line and a small hook at its right end. The sixth and seventh arrows are identical to the fifth, showing a more complex arrow with a tail and a hook.

Handwritten notes on a blackboard:

- Top left: 2
- Top right: $\frac{7}{2} + 1$
- Bottom left: A diagram showing three asterisks ($*$) arranged in a curve, with a bracket underneath them.
- Bottom right: $i, j = (5, 6, 7)$

$$(i = 1 \text{ \&\& } j > \frac{n}{2} + 1)$$

bottom left

$i = n \text{ \& \& } j < \frac{n+1}{2}$

$$\left(\begin{matrix} j = n \\ 2 \\ i = \frac{n}{2} + 1 \end{matrix} \right)$$

middle
row

← Right
bottom

(Top right)



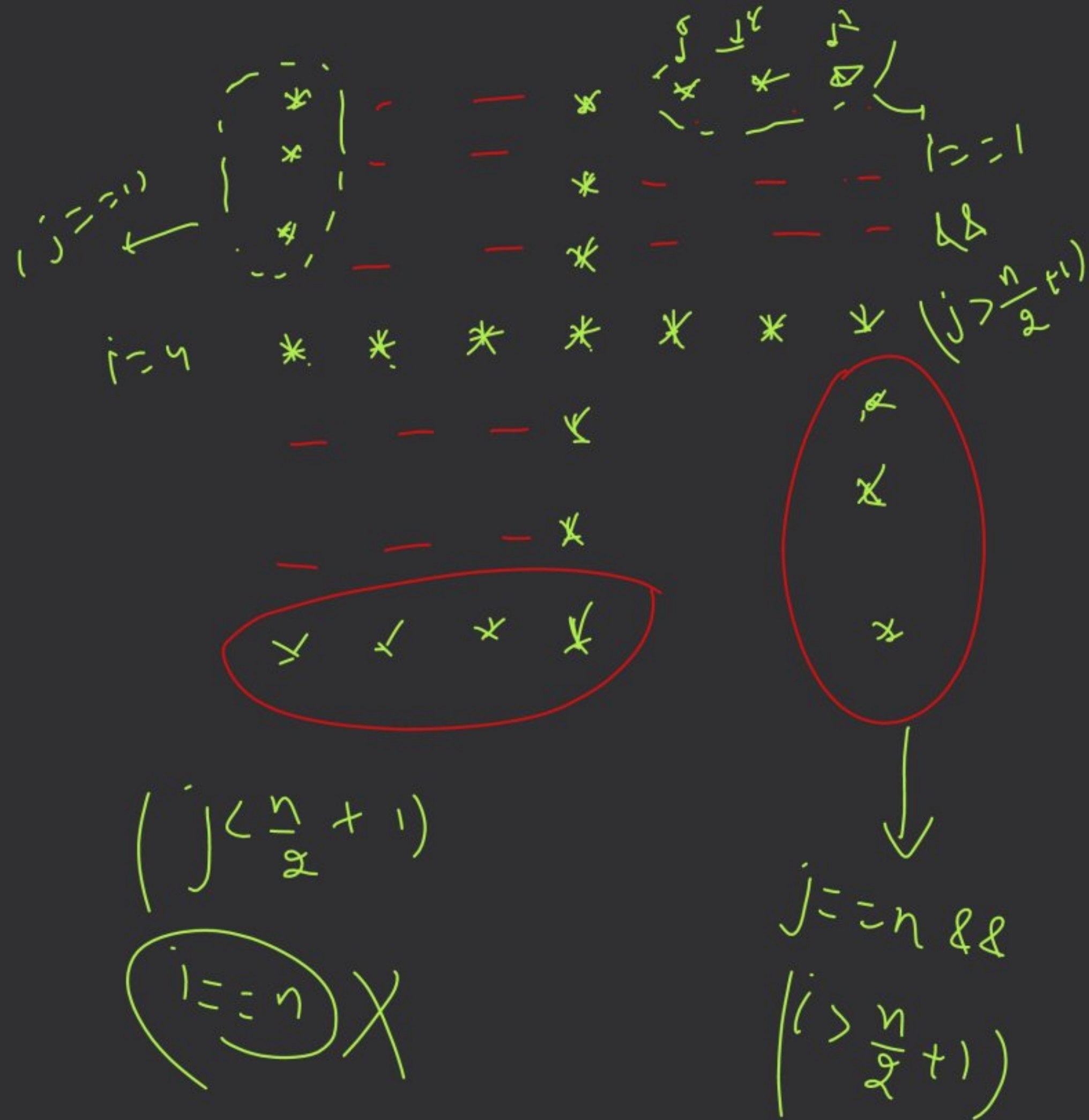
$n=7$
int $i=1$;

while ($i \leq n$) {
 (int $j=1$);
 while ($j \leq n$) {
 String $ch = ""$;
 if ($i == n/2 + 1$)
 $ch = "*" + ch$;
 }

($i=4$)
↓
($j=4, 2, 3, 4, 5, 6, 7$)

else if ($j == n/2 + 1$)
 $ch = ch + "*" \rightarrow TR$
 $cout << ch$;
 $j++$;
else if ($i == 1 \ \&\& \ j > n/2 + 1$) {
 $ch = "*" + ch$;
 $cout << ch$;
 $i++$;
 $j--$;
 $ch = ""$;
 $j = n$;
 else if ($i == n \ \&\& \ j < n/2 + 1$) $\rightarrow BL$
 $ch = "*" + ch$;
 else if ($j == n \ \&\& \ i > n/2 + 1$) $ch = "*" + ch$;
 $j--$;
 $i = 1$;
}

$i=1, 2, 3, 4, 5, 6, 7$
↓
($j=4$)
(BR)



function :-
method



Block of code that perform a specific task.

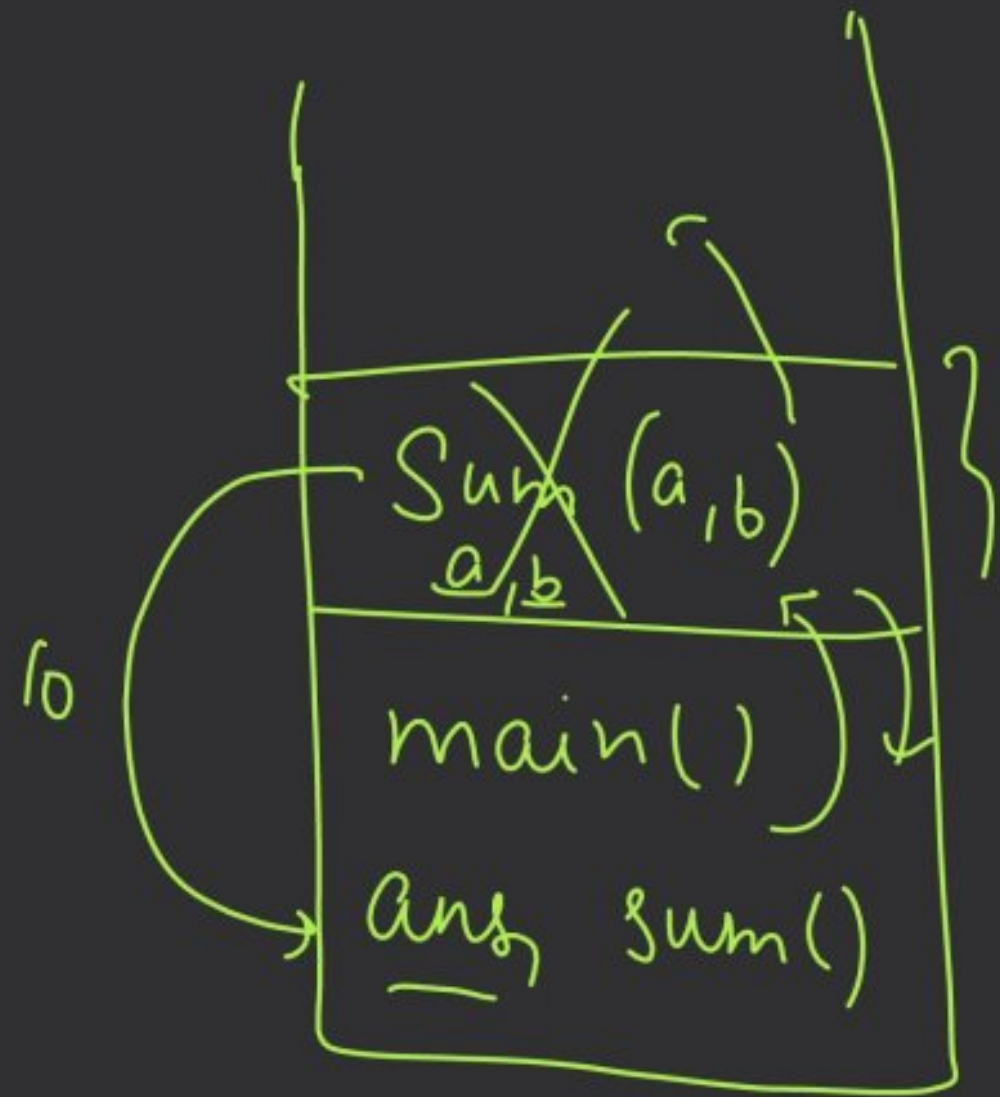
return type fn name () {

```
public class main {  
    public static void greet () {  
        sout ("Hello students");  
    }  
    psvm (String[] args) {  
        greet();  
    }  
}
```

(Last line) →

main()
greet()
main()

public class _____ {



Parameter + Return Type

```
public class Main {  
    qsum(string[] args) {
```

```
        int ans = Sum(3,7);  
        sout(ans);  
    }
```

```
static int Sum(int a, int b) {  
    // return a+b;  
    sout(a);  
    sout(b);  
}
```


Problem 1: average of two number

(for void type)

→ Static void avg(int a, int b) {

int avg = (a+b)/2;

cout << avg;

}

main() {

float ans = avg(5, 8);
cout << ans;

$$\frac{8+5}{2} = \frac{13}{2} = 6.5$$

Static float avg(int a, int b);

float av = (a+b)/2;

return av;

}

(float)(a+b)/2;

Check A number is even or odd →

M.W.
Write a
Jⁿ that take
input and
print the table

psvm (String [] args) {

boolean check_isEven = isEven(25);

System.out (check_isEven);

}

return
 $n \% 2 == 0$;

if ($n \% 2 == 0$)
return true;
else
return false;

public static boolean isEven (int n) {
boolean isEven = _____
System.out (isEven);
}

public static boolean isEven (int n) {

Problem 4: factorial $\rightarrow 5! = 5 \times 4 \times 3 \times 2 \times 1$
 $1 \times 2 \times 3 \times 4 \times 5$

public static int factorial(int n) { $(5! = 4! \times 5) \leftarrow$

int fact = 1

for (int i = 2; i <= n; i++) { $n=5$

fact = fact * i;

}

return fact;

}

fact = ~~1~~ 2
~~6~~ 24
 120

$3! \times 4$
 $2! \times 3$
 $(1! \times 2)$

H.W.
Prime Number

6
 $\nearrow$
 3!

$4! = 3! \times 4$
 $1! \times 3$
 $1!$

static void change (int n)

{
 n = 100;
}

n → 10
100

int n = 10;
Sort(n);

→ Change(n);
Sort(n);

n → (10)

void prime()
(11)


```
1 package methods;
2
3 public class LearnMethods {
4     public static void main(String[] args) {
5         // System.out.println((args[0]));
6         // System.out.println("1");
7         // greet();
8         // System.out.println("5");
9
10        // average(4,6);
11        System.out.println("hello");
12        int a = 4;
13        int b = 6;
14        System.out.println(a+b);
15        System.out.println("hello");
16    }
17 }
18 //
19 // public static void greet(){
20 //     System.out.println("Hello World");
21 //     average(4,6);
22 // }
23 //
24 // public static void average(int a,int b){
25 //     float avg = (float)((a + b) / 2);
```

✓
return;
return a;

int
float
boolean

LearnMethods.java × LearnMath.java Math.java

3 public class LearnMethods {

4 public static void main(String[] args) {

9

10 // average(4,6);

11

12 greet();

13

14 }

15

16 public static void greet(){ 1usage

17 System.out.println("Hello World");

18 average(4,6);

19 }

20

21 public static void average(int a,int b){ 1usage

22 // float avg = (float)((a + b) / 2);

23 int avg = (a+b) / 2;

24 System.out.println("average is " + avg);

25 }

26 }

27

Problems

File 1

Project Errors

Vulnerable Dependencies

Qodana

LearnMethods.java C:\Users\91950\IdeaProjects\thread\src\methods 1 problem

Commented out code (5 lines) :5

thread > src > methods > LearnMethods > average 23:29 CRLF UTF-8 4 spa

Hand-drawn red box containing the text:

- average()
- greet()
- main()

Scanned with OKEN Scanner


```
1 package methods;
2
3 public class LearnMethods {
4     @ public static <string> void main(String[] args) {
5         // System.out.println(args[0]);
6         // System.out.println("1");
7
8         for(String arg : args){
9             System.out.println(arg);
10        }
11        // greet();
12        // System.out.println("5");
13
14        // average(4,6);
15
16        // greet("Gaurav");
17        // int x = 10;
18        // System.out.println(x);
19        // change(x);
20        // System.out.println(x);
21    }
22
23
24    // public static void change(int x){
```

int () arr;

for (string arg : args) {

print (arg);

}

Run LearnMethods ×

is the array of